QPCODE: 25900792 -Scheme

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2025 FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCCFS100 - INTRODUCTION TO DIGITAL FORENSICS AND TOOLS

Part A (Short Answer) Answer any 10 Questions [10x1 = 10]

1. Explain the role of forensic examiner in the judicial system.

The forensic examiner acts as an expert witness, providing unbiased opinions based on training and experience to help the court understand digital evidence. **(1 mark)**

2. Describe Prefetch files.

Prefetch files are created by Windows to accelerate application startup by preloading code and data. They record: Application execution history, Timestamps, Accessed files and directories. **(1 mark)**

3. Compare two common techniques used in anti-forensics to obfuscate data.

Data Wiping: Overwrites data with random patterns to prevent recovery. Encryption: Scrambles data using keys, making it unreadable without decryption. **(1 mark)**

4. Explain Virtual Lab.

A Virtual Lab is a controlled, isolated digital environment using virtualization software to replicate systems for safe forensic analysis, testing tools, and evidence examination without risking the original data. **(1 mark)**

5. Describe Allocated space.

Allocated space is the portion of a storage device actively used by the file system to store file contents and metadata of existing, non-deleted files. **(1 mark)**

6. Summarize the main features of an NTFS file system.

NTFS (New Technology File System) supports: Large volumes, File compression, Encryption (EFS), Journaling via $LogFile, Metadata in $MFT, Access control lists (ACLs) **(1 mark)**

7. Explain analysis phase in digital forencics investigation.

Applying scientific method and critical thinking to interpret evidence and answer who, what, when, where, how, and why. **(1 mark)**

8. Explain the contents that a good Final Report should contain.

A good Final Report should contain: Summary of findings, Methodology used, Evidence examined, Conclusions drawn, Supporting documentation (e.g., logs, hashes, screenshots) **(1 mark)**

9. Summarize the main objectives of SCA.

The Stored Communications Act (SCA) Protect privacy of stored digital communications and records while defining lawful government access procedures based on provider type (ECS/RCS), data category, and required legal process. **(1 mark)**

10. Describe Forensically clean media.

Forensically clean media is storage media that has been wiped (zero-filled or randomized) to ensure no prior data remains, preventing contamination of evidence. **(1 mark)**

11. Explain the importance of photography and notes in documenting the crime scene.

Photography and notes document the crime scene's original state, preserve visual and contextual evidence, and ensure accurate reconstruction for court admissibility. **(1 mark)**

12. Discuss Various types of protocols used to send and receive e-mail.

SMTP (sends email), POP3/IMAP (receives email). **(1 mark)**

Part B (Short Essay) Answer any 4 Questions Each Question [4x5 =20]

13. Explain policies and procedures involved in handling evidence.

Define scope, objectives, roles, and responsibilities of investigators**.** Outline legal/ethical considerations (e.g., ECPA, authorization). Guide evidence collection, preservation, and reporting. Step-by-step collection, analysis, and chain-of-custody maintenance. Use checklists, hashing, secure transport, and verification protocols. **(5 marks)**

14. Explain Hybrid sleep.

Definition: Combines sleep and hibernation modes. RAM Storage: Saves system state in RAM for quick resume. Disk Backup: Writes system state to hard disk to preserve data if power

fails. Power Safety: Prevents data loss during power outage unlike normal sleep. Forensic Relevance: Volatile RAM data may be recovered from hibernation file (hiberfil.sys). **(5 marks)**

15. Illustrate the different types of cellular networks.

Image of a cellular network **(3 marks)**

Explain- Analog voice-only (e.g., AMPS), Digital voice & SMS (e.g., GSM, CDMA), Mobile data & internet (e.g., UMTS, CDMA2000), High-speed data, VoIP (e.g., LTE), Ultra-fast, low latency, IoT support. **(2 marks)**

16. Discuss about live acquisition concerns.

Data Volatility, System Alteration, Order of Volatility, Tool Reliability, Legal & Authorization. **(5 marks)**

17. Explain the uses of Hashing.

Verify Integrity, Chain of Custody, Identify Known Files, Detect Tampering Forensic Imaging. **(5marks)**

18. Discuss the role of mobile forensic tools in recovering data from applications like WhatsApp, Facebook, and Instagram. What types of evidence can these tools extract from messaging apps?

[Award marks to all responses.]

Part C (Essay) Answer any 2 Questions Each Question [2x10=20]

19. Explain the different organization that make significant contributions to the deciplin of digital forensics.

NIST (National Institute of Standards and Technology), SWGDE (Scientific Working Group on Digital Evidence**),** IOCE (International Organization on Computer Evidence), HTCIA (High Technology Crime Investigation Association**),** IACIS (International Association of Computer Investigative Specialists) **(5 marks)**

Organizations ensure standardization, tool reliability, training, and legal admissibility in digital forensic investigations. EnCase, FTK (Forensic Toolkit), Autopsy, X-Ways Forensics, X-Ways Forensics. Forensic imaging, Hashing tools. Explain any tools **(5 marks)**

20. Discuss about Crime scenes and collecting evidence.

Digital vs Physical Crime Scene, Preservation First, Order of Volatility, Evidence Types Collected Volatile, Powered-off imaging, Documentation & Chain of Custody, Legal Considerations. **(5 marks)**

Explained the process of identifying, Collection, preserving, analyzing and presenting digital evidence **(5 marks)**

21. Ilustrate the techniques used in network forensics to analyze network traffic and evaluate the challenges faced in this field.

Network forensics involves capturing, recording, and analyzing network packets to reconstruct events, identify anomalies, and gather evidence. Packet Capture and Analysis (Sniffing), Log File Examination, Traffic Reconstruction, Anomaly Detection, IP Address Tracing – Explain any five points **(5 marks)**

Challenges Faced in Network Forensics: Large Data Volume, Volatility and Evidence Loss, Attacker Evasion, Jurisdictional Issues, Technical Hurdles. **(5 marks)**